Restore the USB connectivity firmware on UNO R4 WiFi with espflash

Source: Arduino Help Center — https://support.arduino.cc/hc/en-us/articles/16379769332892-Restore-the-USB-connectivity-firmware-on-UNO-R4-WiFi-with-espflash (saved 2026-11-01)

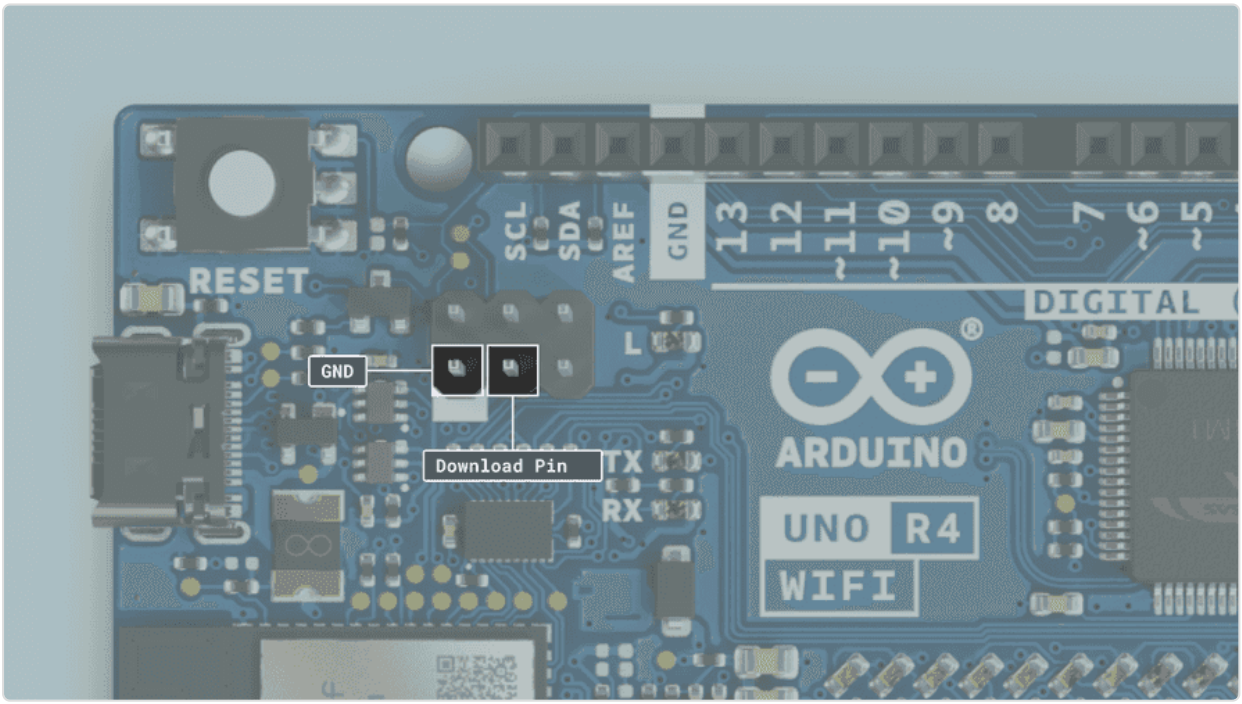
The UNO R4 WiFi has a ESP32-S3 chip, which handles the USB-to-serial communication. The firmware for this chip can normally be upgraded using Arduino IDE or Arduino Cloud, but if this is not possible the firmware can be restored using **espflash**.

You may want to use this procedure if:

- Your UNO R4 WiFi is detected as a generic ESP32 board by Arduino IDE or Arduino Cloud.
- Your UNO R4 WiFi isn't detected as any board by Arduino IDE or Arduino Cloud, despite being connected to your computer with a working data USB cable.

Follow these steps:

1. Unplug UNO R4 WiFi and any non-essential USB devices from your computer.
2. On the board, find the **GND** and **Download** pins on the 6-pin header next to the USB-C connector:



3. Connect the board to your computer while shorting the GND and Download pins.
    - This can easily be done by connecting a female-to-female jumper wire between the two pins.
    - If you don't have a female-to-female jumper wire, you can use some other pointy, conductive object (such as one end of a male-to-male jumper wire) and position to have contact with both pins. You can release the short after you've connected the board to your computer.
4. The ESP32 should now be in Download mode, which allows firmware to be flashed.
    - You may get a system message about a new device being connected. If prompted, allow it to connect.
5. Download and extract the .ZIP file for your system:
    - unor4wifi-update-windows.zip
    - unor4wifi-update-macos.zip
    - unor4wifi-update-linux.zip
6. Open your system's command line application inside the extracted folder.
    - **Windows:** Hold `⇧Shift` and right-click any blank space inside the extracted folder. In the context menu, select **Open command window here / Open PowerShell window here**.
    - **macOS:** Control-click on the extracted `unor4wifi-update-macos` folder and select "**Services > New Terminal at Folder**" from the context menu. A terminal window will open.
    - **Linux:** Open a command line terminal in the extracted folder.
7. Run the command:
    - **Windows:** `bin\espflash write-bin -b 115200 0x0 (Get-Item .\firmware\UNOR4-WIFI-S3-*.bin).FullName`
    - **macOS/Linux:** `./bin/espflash write-bin -b 115200 0x0 firmware/UNOR4-WIFI-S3-*.bin`

> **Note**
>
> If you're using macOS, you may need to adjust your system's security settings to allow this command if you encounter a warning pop-up. In the System Settings app, select *Privacy & Security* and scroll down to the *Security* section. Click *Allow Anyway* to allow espflash to run, and repeat the previous step.

8. The terminal window might now show a list of the serial ports present on your computer. Use the arrow keys on your keyboard to select the port for the board and then press the `Enter` key.
    - **Windows:** The board will have a name like "USB Serial Device".
    - **macOS:** The board will be named "USB JTAG_serial debug unit". Two ports with this name will be available, select the port that starts with `/dev/cu.usbmodem*`.
    - If you can't find it, try following steps 1–3 again.
9. The terminal window may show the following prompt:

    ```
    ? Remember this serial port for future use? (y/n) ›
    ```

    Press the `N` key on your keyboard.
10. The flashing process should now start. Wait for it to finish, as indicated by output that looks something like this:

    ```
    [2023-07-11T08:59:44Z INFO ] Connecting...
    [2023-07-11T08:59:45Z INFO ] Using flash stub
    Chip type:         esp32s3 (revision v0.1)
    Crystal frequency: 40MHz
    Flash size:        8MB
    Features:          WiFi, BLE
    MAC address:       dc:54:75:c4:c6:54
    [00:00:14] [========================================]     689/689     0x0
    ```

11. Close the terminal window.
12. Disconnect the USB cable of the **UNO R4 WiFi** board from your computer.
13. Ensure that the Download and GND pins are no longer shorted.
14. Connect the **UNO R4 WiFi** board to your computer with the USB cable again.